The given determinant is in cyclic symmetric form in x, y and z. Therefore $(y - z)$ and $(z - x)$ are also factors.

The degree of the product of the factors $(x-y)(y-z)(z-x)$ is 3 and the degree of the product of the leading diagonal elements $1\times y^2\times z^3$ is 5.

Therefore the other factor is $k(x^2+y^2+z^2)+\ell(xy+yz+zx)$.

Thus $$\begin{vmatrix} 1 & x^2 & x^3 \\ 1 & y^2 & y^3 \\ 1 & z^2 & z^3 \end{vmatrix} = [k(x^2+y^2+z^2)+\ell(xy+yz+zx)]\times(x-y)(y-z)(z-x)\ .$$

Putting $x = 0,\ y = 1$ and $z = 2$, we get

$$\begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 4 & 8 \end{vmatrix} = \big[k(0+1+4)+\ell(0+2+0)\big](-1)(1-2)(2-0)$$

$$\Rightarrow \qquad (8-4) = [(5k+2\ell)](-1)(-1)(2)$$

$$4 = 10k+4\ell \Rightarrow 5k+2\ell = 2. \qquad \text{... (1)}$$

Putting $x = 0, y = -1$ and $z = 1$, We get

$$\begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{vmatrix} = [k(2)+\ell(-1)](1)(-2)(1)$$

$$\Rightarrow \quad [(2k-\ell)(-2)] = 2$$

$$2k-\ell = -1. \qquad \text{... (2)}$$

Solving (1) and (2), we get $k = 0,\ \ell = 1$.

Therefore $$\begin{vmatrix} 1 & x^2 & x^3 \\ 1 & y^2 & y^3 \\ 1 & z^2 & z^3 \end{vmatrix} = (x-y)(y-z)(z-x)(xy+yz+zx).$$

Example 7.25

Prove that $|A| = \begin{vmatrix} (q+r)^2 & p^2 & p^2 \\ q^2 & (r+p)^2 & q^2 \\ r^2 & r^2 & (p+q)^2 \end{vmatrix} = 2pqr\ (p+q+r)^3.$

Solution :

Taking $p = 0$, we get $|A| = \begin{vmatrix} (q+r)^2 & 0 & 0 \\ q^2 & r^2 & q^2 \\ r^2 & r^2 & q^2 \end{vmatrix} = 0.$

Therefore, $(p - 0)$ is a factor. That is, p is a factor.

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